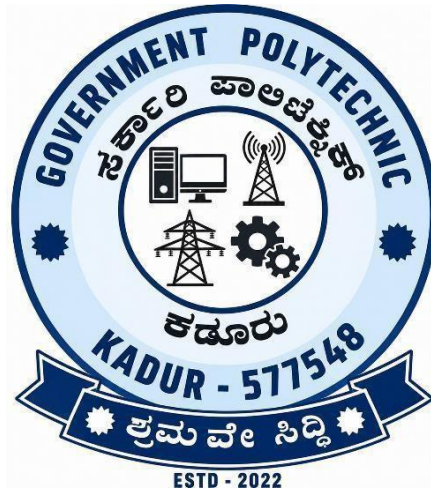


# GOVERNMENT POLYTECHNIC, KADUR

## COMPUTER SCIENCE AND ENGINEERING

### ONLINE COURSE REPORT ON:

1. INTRODUCTION OF ARTIFICIAL INTELLIGENCE
2. STATISTICS ESSENTIALS WITH PYTHON



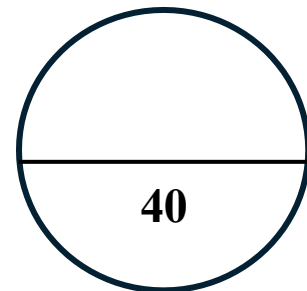
SUBMITTED BY : VASUDEVAN

OBTAINED MARKS:

REGISTER NO : 197CS24045

SEMISTER : 5<sup>th</sup> SEM 3<sup>rd</sup> YEAR

SUBJECT : FULL STACK DEVELOPMENT



Signature:

## **INFOSYS SPRINGBOARD**

Infosys Springboard is a free digital learning platform developed by Infosys to provide quality education and industry-relevant skills to students, teachers, and professionals. The platform offers self-paced learning through video lectures, reading materials, practical exercises, and assessments. It helps learners enhance their technical knowledge and prepare for future careers.



### **Key Features:**

1. Free access to a wide range of courses.
2. Self-paced online learning.
3. Video lectures and study materials.
4. Quizzes and assessments.
5. Digital certificates after course completion.
6. Industry-focused learning paths.
7. Progress tracking to monitor learning achievements.
8. Mobile-friendly access for learning anytime, anywhere.

## **Benefits Of Infosys Springboard:**

One of the popular courses on the platform is Introduction to Artificial Intelligence, which covers AI fundamentals, Machine Learning, Deep Learning, AI business applications, and AI platforms such as Infosys NIA. The course helps learners understand how AI is transforming industries like healthcare, banking, retail, manufacturing, and education.



## **Improves technical and professional skills:**

1. Enhances problem-solving and analytical abilities.
2. Supports career development and job readiness.
3. Provides learning opportunities for students and working professionals.
4. Encourages continuous learning and innovation.
5. Improves communication and teamwork skills through collaboration on projects and professional interactions.
6. Builds confidence in using modern tools and technologies relevant to the industry.

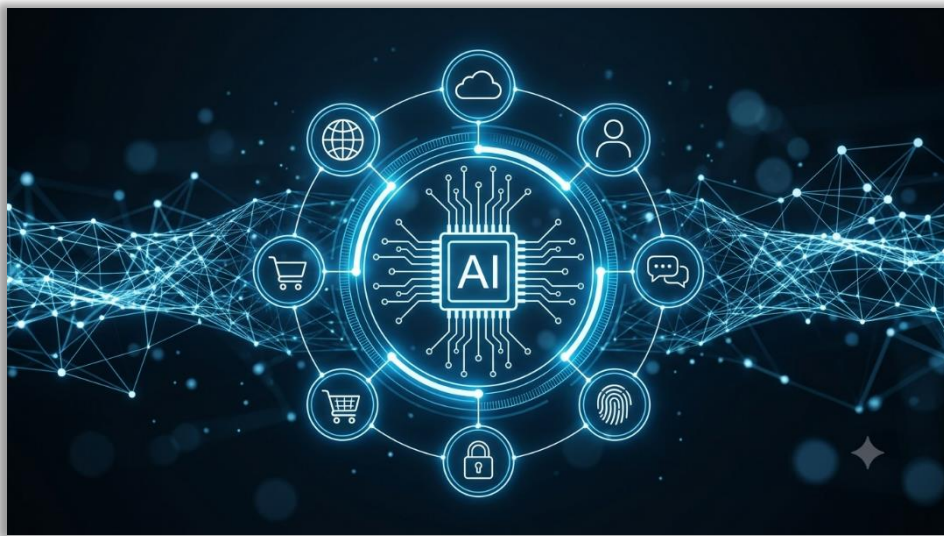
| SL NO | COURSE NAME                             | TIME   |
|-------|-----------------------------------------|--------|
| 1     | INTRODUCTION OF ARTIFICIAL INTELLIGENCE | 6H 18M |
| 2     | STATISTICS ESSENTIALS WITH PYTHON       | 4H 33M |



## ARTIFICIAL INTELLIGENCE

### **Introduction to Artificial Intelligence:**

Artificial Intelligence (AI) is the branch of computer science that develops systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, language understanding, decision-making and pattern recognition. AI improves productivity, automates repetitive work and helps organizations make better decisions.



### **Learning Outcomes:**

7. Understand AI concepts.
8. Identify AI applications.
9. Explain machine learning basics.
10. Recognize business value of AI.

### **Why AI?**

AI increases efficiency, reduces errors, supports faster decisions, improves customer experience and enables innovation across industries. AI also enables 24/7 availability, allowing systems to operate continuously without fatigue or downtime.

## Machine Learning Basics

1. **Machine Learning (ML):** Subset of AI that learns patterns from data to improve performance without explicit programming.
2. **How Machines Predict:** Models learn patterns from historical (training) data, then apply that pattern to new, unseen data to estimate outcomes.
3. **Supervised Learning:** Trains on labeled data to predict outcomes — divided into Classification (predicts categories) and Regression (predicts continuous values).
4. **Classification Algorithms:** SVM (finds best separating boundary), Decision Tree (flowchart-based splits), KNN (classifies by nearest neighbors), Logistic Regression (predicts binary outcomes like yes/no).
5. **Unsupervised Learning Types:** Clustering (groups similar data points) and Association (finds relationships between variables in large datasets).
6. **Deep Learning:** Uses multi-layered artificial neural networks to automatically learn complex patterns, widely used in image recognition and NLP.
7. **Types of ANN:** Feedforward Neural Networks (basic structure), CNN (best for image data), RNN (best for sequential/time-based data).
8. **Bias-Variance Tradeoff:** Balances model simplicity (bias) and sensitivity to training data (variance) to avoid overfitting or underfitting.
9. **Timeseries Forecasting:** Predicts future values based on time-ordered historical data — used in stock prices, weather, and sales forecasting.
10. **AI is the New UI:** AI-powered interfaces (chatbots, voice assistants) are becoming the primary way users interact with technology, replacing traditional buttons and menus



## WHAT I LEARNED ON THIS COURSE:

1. AI vs ML vs DL — AI is the broad concept; ML is a subset that learns from data; DL is a subset of ML using neural networks.
2. Supervised Learning — Trains models on labelled data to predict outcomes (e.g., spam detection).
3. Generative AI — Creates new content (text, images, code) using models like ChatGPT and DALL·E.
4. Bias in AI — Models can inherit bias from training data, leading to unfair outcomes.
5. Advantage — Increases speed and accuracy of decision-making.
6. Disadvantage — High implementation cost and risk of over-dependence on technology.
7. Career Scope — Growing demand for AI Engineers, Data Scientists, and ML Researchers.

## Conclusion:

Mastering AI vs ML vs DL, ethical practices, and generative AI trends is essential for building future-ready skills and driving responsible innovation across industries. Organizations that adopt AI early gain a competitive edge through smarter, faster, and more efficient operations, while individuals who understand these concepts will be better prepared for future careers.



# STATISTICS ESSENTIALS WITH PYTHON

## Introduction to Data Science:

Data Science collects, processes, analyses, and interprets data to discover meaningful insights. It combines statistics, programming, machine learning, and data visualization to solve real-world problems and support data-driven decision-making, widely used across healthcare, finance, education, and business.



1. Uses programming languages such as Python and R
2. Helps organizations make better decisions by identifying patterns and predicting trends

## Basic Concept of Statistics:

Statistics is the branch of mathematics focused on collecting, organizing, analysing, interpreting, and presenting data. It is divided into **Descriptive Statistics** and **Inferential Statistics**.

3. Measures of central tendency: Mean, Median, Mode
4. Measures of dispersion: Range, Variance, Standard Deviation

## Calculating Mode:

Mode is the value occurring most frequently in a dataset — applies to numerical or categorical data. A dataset can be unimodal, bimodal, multimodal, or have no mode.

|                                                         |
|---------------------------------------------------------|
| <b>Formula:</b> Mode = Value with the highest frequency |
|---------------------------------------------------------|



## Calculating Dispersion Measures:

Dispersion measures show how spread out values are around the central value. Lower dispersion = values close together; higher dispersion = values widely spread.

**Formula:**

Range = Max – Min | Variance = Avg. of squared deviations from mean | SD =  $\sqrt{\text{Variance}}$

## Histogram:

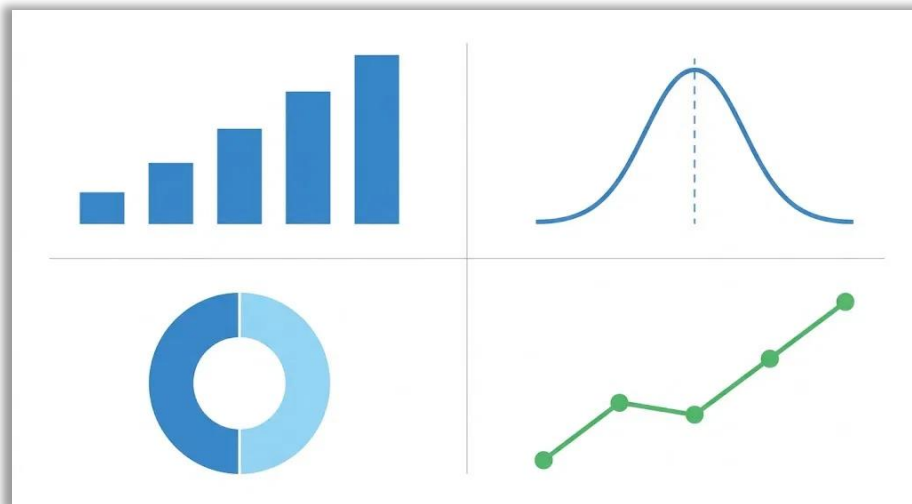
A histogram graphs the frequency distribution of numerical data by grouping values into bins, revealing shape, spread, and outliers.

**Formula:** Frequency = Number of observations in each bin

## Correlation Function:

Measures the strength and direction of the relationship between two variables. Ranges from -1 to +1 (+1 = perfect positive, -1 = perfect negative, 0 = no relationship).

**Formula:**  $r = \text{Cov}(X, Y) / (\sigma X \times \sigma Y)$



## Pandas Data Frame:

A two-dimensional data structure in Python used to store and organize data in rows/columns — used for cleaning, filtering, sorting, and visualization.

## NumPy Package

A Python library for numerical computing, supporting multi-dimensional arrays and fast mathematical operations — performs much faster than standard Python lists.

## Hypothesis Testing

Determines whether a claim about a population is supported by sample data, testing the Null Hypothesis ( $H_0$ ) against the Alternative Hypothesis ( $H_1$ ).

|                                                                              |
|------------------------------------------------------------------------------|
| <b>Formula:</b> $P(A \text{ or } B) = P(A) + P(B)$ (when mutually exclusive) |
|------------------------------------------------------------------------------|

## Finding the Measurement & Denominator

The process of calculating key statistical values (mean, variance, probability) to summarize and understand a dataset. The denominator represents the total number of values/observations used in these formulas.

|                                                                                            |
|--------------------------------------------------------------------------------------------|
| <b>Formula:</b> Mean = $\Sigma X / n$   Probability = Favourable outcomes / Total outcomes |
|--------------------------------------------------------------------------------------------|

## What You Will Learn

1. Data Science fundamentals and applications
2. Mean, Median, Mode, and dispersion measures
3. Histograms, Correlation, Hypothesis Testing, and OLS Regression
4. Using Pandas and NumPy for real-world data analysis in Python
5. Understanding Correlation and Regression for predictive analysis
6. Applying Hypothesis Testing to validate data-driven decisions
7. Building and manipulating Pandas Data Frames for data handling

## Conclusion:

Data Science combines statistics, programming, and machine learning to extract meaningful insights from data. Techniques like descriptive statistics, hypothesis testing, correlation, and regression — along with Python libraries like NumPy and Pandas — enable accurate predictions and better decision-making. As data volumes grow, Data Science remains essential for innovation across healthcare, finance, education, and business.